

SOC Alert Triage at Scale

Splunk BOTS v3 Dataset — Multi-Source Security Investigation

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Dataset	Splunk BOTS v3 (Boss of the SOC) — Frothly Inc. simulated environment
Platform	Splunk Enterprise
Report Date	July 2026
Log Sources Reviewed	Linux bash_history, Apache access logs, AWS CloudTrail, AWS GuardDuty, Windows Security Events, DNS query logs
Findings Documented	4 (2 True Positive, 1 False Positive, 1 Clean Sweep)

Executive Summary

This report documents a large-scale alert triage exercise conducted using Splunk's official BOTS v3 dataset - a realistic, pre-recorded collection of security telemetry from a simulated company ("Frothly Inc.") spanning Linux hosts, AWS cloud infrastructure, and Windows endpoints. Unlike a self-built lab where the analyst controls and therefore already knows the outcome of every event, this exercise required investigating unfamiliar, ambiguous data - mirroring the real daily workflow of a SOC L1 analyst. Each finding below documents not only what was found, but the reasoning process used to correctly classify it, including cases where an initially suspicious-looking event was ultimately ruled out as benign after deeper investigation.

Finding 1 — IoT Botnet Command Injection Attempt (Hakai)

Log Source	Apache access_combined (/var/log/httpd/access_log)
Source IP	156.202.170.156
Payload Host	185.172.164.41
Target Endpoint	/login.cgi
Timestamp	Aug 20, 2018, 13:31:22 UTC
Severity	High
Classification	True Positive (Attack Attempt — Failed)
Analysis	A single HTTP request targeting /login.cgi contained a command injection payload attempting to download and execute a remote file via wget and sh, consistent with known IoT botnet infection techniques. The User-Agent explicitly identified itself as Hakai/2.0, a known malware family that targets routers and IoT devices via command injection in login forms. The web server responded with 400 Bad Request, indicating the malformed request was rejected before execution - the attack did not succeed.
Recommended Action	Block source IP and payload host at firewall/WAF level. Verify /login.cgi (if present) is patched against command injection. Add WAF detection rule for the Hakai User-Agent and wget/sh injection patterns.

Splunk Query Used

```
index=* host="ubuntu-victim" source="/var/log/httpd/access_log" "Hakai"
| table _time, _raw
```

Finding 2 — Bash History Review: User klagerfield (False Positive)

Log Source	Linux bash_history (/home/klagerfield/.bash_history), host: hoth
User	klagerfield
Timestamp Range	Aug 20, 2018, 17:05 - 18:58 UTC
Severity	Low (initially flagged, downgraded after review)
Classification	False Positive / Benign

Analysis	Initial review raised concern due to use of sudo bash and access to /etc/passwd, commands commonly associated with reconnaissance following privilege escalation. Deeper analysis of the full, correctly time-ordered command sequence showed this was routine Splunk application administration - editing inputs.conf and navigating standard local/system config directories. No access to /etc/shadow, no credential harvesting, and no lateral movement was observed. A separate, unrelated check of the Tomcat service (ps -aux grep tomcat) occurred nearly two hours later with no follow-up exploitation activity. Note: bash_history timestamps are written per shell session rather than per command, limiting precise sequencing of rapid actions.
Recommended Action	No escalation required. Close as benign. If this pattern recurs, correlate with authentication logs to confirm the session ties to legitimate admin duties.

Splunk Query Used

```
index=botsv3 earliest=0 source="/home/klagerfield/.bash_history" host="hoth"
| sort _time
| table _time, _raw
```

Finding 3 — AWS GuardDuty: Port Probe on Production Web Server

Log Source	AWS GuardDuty Finding (via CloudTrail/EventBridge)
Finding Type	Recon:EC2/PortProbeUnprotectedPort
Target	EC2 instance i-0cc93bade2b3cba63 (production-FrothlyWebPubSecGroup, WebServers ASG)
Probed Port	22 (SSH)
Source IP	122.112.243.11 (Beijing, China — China Telecom)
Threat Intel	Flagged as 'Scanner' on ProofPoint threat intelligence list
GuardDuty Severity	2 (Low-Medium)
Classification	True Positive — Reconnaissance
Analysis	AWS GuardDuty detected an external IP address, already identified on a recognized threat intelligence feed as a known scanner, probing SSH (port 22) on a production web server. The probe reached the instance (blocked: false), indicating detection occurred but the connection was not stopped at the network layer. No evidence of a successful login or follow-up exploitation was found. This represents the reconnaissance phase of a potential attack rather than a confirmed breach.

Recommended Action	Restrict security group rules so SSH (port 22) is reachable only from trusted IP ranges (VPN/bastion host), not the public internet. Block the source IP at the network ACL level. Enforce SSH key-based authentication only. Route GuardDuty findings to a real-time alerting channel.
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Finding 4 — DNS Traffic Sweep (Clean Result)

Log Source	DNS query logs (stream:dns)
Scope Reviewed	Top 30 most frequent DNS queries across the environment
Classification	Clean — No Indicators of Compromise Found
Analysis	A review of the highest-volume DNS queries showed exclusively expected, legitimate destinations: internal Splunk infrastructure (splunk.froth.ly), AWS-managed services (RDS, SSM), Microsoft 365 services (Outlook, login, OneDrive/Office apps), Windows telemetry, and standard web/advertising domains (Google, Skype, Edge). No unusually long or randomized domain names, no excessive NXDOMAIN patterns, and no known command-and-control or DNS tunneling indicators were identified.
Recommended Action	No action required. Recommend periodic re-baselining of top DNS queries to maintain an accurate 'known-good' reference for future anomaly detection.

Splunk Query Used

```
index=botsv3 earliest=0 sourcetype="stream:dns"
| stats count by query
| sort -count
```

Conclusion

This exercise applied the same detection and reasoning skills developed in a controlled home lab (Project 1) to unfamiliar, realistic data at volume. Two genuine security events were identified and correctly classified as true positives (a failed IoT botnet exploitation attempt and an AWS-detected reconnaissance probe against production infrastructure), while two initially suspicious-looking leads were investigated thoroughly and correctly ruled out as benign or clean. This balance - catching real threats while avoiding false escalation - reflects the core judgment skill required in day-to-day SOC L1 alert triage.